Part-3

Determinants

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⇒ For this chapter, all Matrices will be Square Matrix.

⇒ If A is a Matrix then similar to Rank(A), Det(A) is also a Scalar.

★ Some properties of Determinants:

→ Determinant of any $n \times n$ Identity Matrix is 1.

Ex: $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ ↳ (1 → Diagonal, 0 → Everywhere)

→ Determinant changes sign when two rows (or columns) are exchanged.

Ex: $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ So $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$

→ Linearity for one row at a time.

IMP.

Ex. $\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & tb \\ c & td \end{vmatrix}$

$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$

$\begin{vmatrix} ta & tb \\ tc & td \end{vmatrix} = t^2\begin{vmatrix} a & b \\ c & d \end{vmatrix}$

(Use above properties only for below questions)

Que T/F?

$\begin{vmatrix} a+a' & b+b' \\ c+c' & d+d' \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c' & d' \end{vmatrix}$

False

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$\det(A) \Leftrightarrow |A|$

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Qu T/F?

$$\begin{vmatrix} 4 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 4 \begin{vmatrix} 1 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

True
(Property-3)

Qu T/F?

$$\begin{vmatrix} 4 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 4 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 8 & 8 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

True
(Property-3)

Qu T/F?

If two rows of A are equal, then $\det(A) = 0$

$$\begin{vmatrix} a & b \\ a & b \end{vmatrix} \Rightarrow a \begin{vmatrix} 1 & b \\ 1 & b \end{vmatrix} \Rightarrow ab \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}$$

& $ab * 0 \Rightarrow 0$

True

Imp.
Qu T/F? Subtracting a multiple of one row from another row leaves det (A) unchanged.

$$\begin{vmatrix} a & b \\ c-la & d-lb \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \begin{vmatrix} a & b \\ la & lb \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} - l \underbrace{\begin{vmatrix} a & b \\ a & b \end{vmatrix}}_{\hookrightarrow 0}$$

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$ True

So $R_i \rightarrow R_i + kR_j$ will not change determinant.

Que. T/F A matrix with a zero row has det = 0.

$$\Rightarrow \begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} = 0$$ ~~If we swap rows~~

$\downarrow R_1 \rightarrow R_1 + R_2$

$$\begin{vmatrix} c & d \\ c & d \end{vmatrix} = 0$$

True (Many other ways to prove)

Que: T/F Determinant of diagonal Matrix is product of diagonal elements.

$$\Rightarrow \begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & \cdots & \cdots & a_{nn} \end{vmatrix} = (a_{11})(a_{22}) \ldots (a_{nn})$$

$$\Rightarrow (a_{11})(a_{22}) \ldots (a_{nn}) \underbrace{\begin{vmatrix} 1 & 0 & \cdots & 0 \\ \vdots & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & \cdots & 1 \end{vmatrix}}_{\hookrightarrow \text{identity Matrix} = 1}$$

True

Que. T/F? Determinant of upper | lower Matrix is product of diagonal elements

Upper Triangle Matrix

$$\Rightarrow \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{vmatrix} = (a_{11})(a_{22})(a_{33})$$

using $R_i \to R_i + kR_j$

$$\begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} = (a_{11})(a_{22})(a_{33})$$

True

Que. Suppose

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ | & & & \\ | & & & \\ a_{41} & - & - & a_{44} \end{bmatrix}, \quad B = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14}+3a_{13} \\ | & | & | & a_{24}+3a_{23} \\ | & | & | & a_{34}+3a_{33} \\ a_{41} & a_{42} & a_{43} & a_{44}+3a_{43} \end{bmatrix}$$

if $\det(A) = 3$, then $\det(B) = ?$

$\Rightarrow$ Using Property 3

$$|B| = |A| + 3\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{13} \\ a_{21} & | & | & a_{23} \\ | & | & | & a_{33} \\ a_{41} & a_{42} & a_{43} & a_{43} \end{vmatrix}$$

$\hookrightarrow$ 0 as Col 3 & Col 4 are same

So $|B| = |A| = 3$

Or directly in B $C_4 \to C_4 + 3C_3$
which make det unchanged (Already Proved)
So equal to that of A, i.e. 3

Que: Suppose det(A) is 44 & B is obtained by ex. R_2 & R_4 of A then det(B) is?

$\Rightarrow$ -44 (Property 2)

Que: Let det(A) is 44 & a_1, a_2, a_3, a_4 are rows of A. If B is obtained by replacing a_3 by $3a_1 + a_3$, then det(B) is?

$\Rightarrow$ 44 (No Change)

Que: Same as above but by replacing a_3 by $3a_1 + 7a_3$ then det(B) is?

$\Rightarrow$ 7×44

Que: Same as above but by replacing row a_3 by $4a_1 + 5a_2$, then det(B) is?

$\Rightarrow$ 0

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}, \text{ then } B = \begin{bmatrix} a_1 \\ a_2 \\ 4a_1 + 5a_2 \\ a_4 \end{bmatrix} \xrightarrow{a_3 \to a_3 - 4a_1} \begin{bmatrix} a_1 \\ a_2 \\ 5a_2 \\ a_4 \end{bmatrix}$$

$$5 \begin{bmatrix} a_1 \\ a_2 \\ a_2 \\ a_4 \end{bmatrix} \Rightarrow 5 * 0 = 0$$

$\hookrightarrow$ 0 as same rows.

[Also can be done by splitting (Property 3)].

So if one row/col is L.C. of other rows then determinant will be zero.

Que: T/F? If $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ & $B = \begin{bmatrix} a & -c & b \\ d & -f & e \\ g & -i & h \end{bmatrix}$

then $\det(A) = \det(B)$

$\Rightarrow$ True

Imp. Base

Que: Calculate determinant of A by using prev. properties

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix}$$

$$= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}$$

$$= 0 + ad - \begin{vmatrix} b & 0 \\ 0 & c \end{vmatrix} + 0$$

$$= ad - bc$$

Ans.

Similarly, we can do for 3x3 Matrix.

 – 27 Matrix & 6 will survive

↳ Each with only one ele per row.

6 will only survive in 3x3 Matrix because if in any row & col we have atleast one nonzero then it will survive.

So. In row 1 we have → 3 choices (any col non-zero)

In row 2 → 2 choices (other than above col)

In row 3 → 1 choice.

So total = $3 \times 2 \times 1$

= $\underline{\underline{6}}$

☆ So in $n \times n$ matrix expansion, we will have $n!$ matrix, that will survive. (total → n^n)

In 3x3, Expansion, Matrix, final ans. will be

(Refer Lecture: 7a 1:20 - 1:30)

$$\begin{bmatrix} \cdot & & \\ & \cdot & \\ & & \cdot \end{bmatrix} + \begin{bmatrix} \cdot & & \\ & & \cdot \\ & \cdot & \end{bmatrix} + \begin{bmatrix} & \cdot & \\ \cdot & & \\ & & \cdot \end{bmatrix} + \begin{bmatrix} & \cdot & \\ & & \cdot \\ \cdot & & \end{bmatrix}$$

$$+ \begin{bmatrix} & & \cdot \\ & \cdot & \\ \cdot & & \end{bmatrix} + \begin{bmatrix} & & \cdot \\ \cdot & & \\ & \cdot & \end{bmatrix}$$

Imp.

☆ i) $a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}$ - -

Here,

$a_{1x}a_{2y}a_{3z}$ is pattern and x, y & z are permutation of 1, 2 & 3 & sign depends on if we need to exchange rows to make matrix diagonal or not.

Ques: Let A be an 4x4 Matrix. In the expression of the det (A) what is the sign of the product $a_{13}\, a_{22}\, a_{34}\, a_{41}$

→ $$\begin{vmatrix} & & \cdot & \\ & \cdot & & \\ & & & \cdot \\ \cdot & & & \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{vmatrix} \cdot & & & \\ & \cdot & & \\ & & & \cdot \\ & & \cdot & \end{vmatrix}$$

$$\xrightarrow{R_3 \leftrightarrow R_4} \begin{vmatrix} \cdot & & & \\ & \cdot & & \\ & & \cdot & \\ & & & \cdot \end{vmatrix}$$

As two swaps are needed $-1 \times -1 = 1$ So positive sign.

pr.

★ $$\begin{vmatrix} & \cdot \\ & \end{vmatrix}_{n \times n} = \sum_{\substack{\text{all} \\ \text{Permutation}}} \pm\, a_{1x}\, a_{2y} \cdots a_{nz}$$

↳ $n!$ terms

} Big formula

$R_i \rightarrow kR_i + xR_j \rightarrow$ Makes $k*\det(A)$

(Refer 7b starting for Revision as Hard to pen the notes)

Que: Solve:

$$\begin{vmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \Rightarrow - a_{13}a_{22}a_{31}a_{44} \quad \cancel{a_{14}a_{22}a_{31}a_{44}} \quad \cancel{a_{14}a_{23}a_{31}a_{44}} + a_{14}a_{23}a_{32}a_{41}$$

$\Rightarrow -1 + 1$

$\Rightarrow 0$

Ans

☆ Three ~~mettus~~ ways to calculate determinant:

→ Using permutation (As done above)

→ Convert given matrix to echelon form (which makes it triangular) & take product of pivots.

→ Calculate using cofactors.

☆ **The Pivot Formula:** After converting A to upper triangular matrix, the pivots $d_1, d_2, \ldots, d_n$ are on diagonal.

If no row exchanges are involved, multiply those pivots to find the determinant.

$$\det(A) = d_1 d_2 \ldots d_n$$

Ex: $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} - \begin{vmatrix} 4 & 5 & 6 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{vmatrix} \Rightarrow -(4 \times 2 \times 1)$

$= -8$

Ans.

→ One thing I can infer is if two $\overset{\text{col}}{\cancel{\text{rows}}}$ are LD, then one zero row & one diagonal will be zero So det. is also zero.

$\Delta \leftrightarrow$ determinant

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☆ **The Cofactor Formulae:** The determinant is the dot product of any row (or column) i of A with its cofactors using other rows.

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} - - + a_{1n}C_{1n}$$

☆ How to find cofactor?

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Sign

$$\begin{vmatrix} + & - & + \\ - & + & - \\ + & - & + \end{vmatrix}$$

Cofactor of $a = \begin{vmatrix} e & f \\ h & i \end{vmatrix}$, $b = -\begin{vmatrix} d & f \\ g & i \end{vmatrix}$

Sign of Cofactor of $a_{ij} = (-1)^{i+j}$ i.e $\begin{cases} \text{odd} \to -1 \\ \text{even} \to +1 \end{cases}$

Que: If in A, A_{ij} is cofactor of a_{ij}, then Δ ($\det(A)$) is given by:

A) $a_{11}A_{21} + a_{12}A_{32} + a_{13}A_{33}$

B) $a_{11}A_{11} + A_{12}a_{21} + A_{13}a_{31}$

C) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$

D) $a_{11}A_{11} + a_{12}A_{12} + a_{13}A_{13}$

Cofactor without symbol is called minor

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Cofactor$(A_{ij}) = (-1)^{i+j} \left| \quad \right|$, minor$(A_{ij}) = \left| \quad \right|$

Que: T/F? If we change a_{11} then cofactor of a_{11} changes?

⇒ **False** C_{ij} don't depend on a_{ij}

Que: T/F In both of the given matrix, C_{11}, C_{12} & C_{13} are same?

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix} \qquad \begin{bmatrix} 4 & 2 & 4 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{bmatrix}$$

⇒ **True**

Que: What could be the corresponding matrices for given determinants? (C_{ij} is cofactor of a_{ij})

a) $D = a_{22}C_{12} + a_{23}C_{22} + a_{21}C_{32}$

⇒ $$\begin{bmatrix} a_{11} & a_{22} & a_{13} \\ a_{21} & a_{23} & a_{23} \\ a_{31} & a_{21} & a_{33} \end{bmatrix}$$

b) $\Delta = a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13}$

$$\begin{bmatrix} a_{21} & a_{22} & a_{23} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Also, it will be zero as two rows are same. (Question can also be asked in this way?)

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IMP

(MSQ)

Que: Which of the following values evaluate to zero?

A) $\Delta = a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13}$
⇒ As two rows (R_1 & R_2) will be same.

B) $\Delta = a_{11}C_{12} + a_{21}C_{22} + a_{31}C_{32}$
⇒ ~~As two cols (C_1 & C_2) will be same.~~ ✓

C) $\Delta = a_{22}C_{11} + a_{21}C_{12} + a_{23}C_{13}$
⇒ No

D) $\Delta = a_{31}C_{11} + a_{32}C_{12} + a_{33}C_{13}$
⇒ As R_1 & R_3 will be same.

*Imp. *confusing

(76 → 55:00)

Que: T/F? If elements of a row or (column) are multiplied with cofactors of any other row (or col), then their sum is zero?

Ex: $a_{21}C_{11} + a_{22}C_{12} + a_{23}C_{13} = 0$?

⇒ ~~False~~ True, because if we multiply then that row gets copied and so zero as two same rows are found.

Imp. Der.

Que: Find product of below matrices:

$$\underset{A}{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}} \cdot \underset{B}{\begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}}$$

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As R_1 with its cofactor — As R_1 with cofactor of R_2 — " "

$$\Rightarrow \quad A \cdot B = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} \checkmark = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Also.

$$B \cdot A = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix} = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↳ Not $|A|^3$ bcz we need matrix operation, we are not finding det. of above matrix

So.

$$\boxed{A \cdot \left(\frac{1}{|A|} B\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$$

$\Rightarrow$ As inverse of A (A^{-1}) is the matrix we multiply with A to get identity matrix.

So. $\boxed{A^{-1} = \dfrac{B}{|A|}}$

So,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}^{-1} = \frac{1}{|A|} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix}$$

So inverse of A is $\boxed{\dfrac{1}{|A|} \times (\text{cofactor matrix of } A)^T}$

☆ Transpose of cofactor of matrix is also called adjoint of matrix.

So $\text{Adj}(A) = [\text{Cof}(A)]^T$

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Rx:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

★ **Imp/Tricky**

Que: Let $A = \begin{bmatrix} -2 & -7 & -9 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}$. Find the third col of A^{-1} without computing other columns.

$\Rightarrow$ Let x (1x3) be the third col of A^{-1}

then $x = A^{-1}\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ → As III col. is L.C. of this le A^{-1}

So $Ax = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

As we already know to solve $Ax = B$ as:

Aug. Matrix →
$$\left[\begin{array}{ccc|c} -2 & -7 & -9 & 0 \\ 2 & 5 & 6 & 0 \\ 1 & 3 & 4 & 1 \end{array}\right] \xleftrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 2 & 5 & 6 & 0 \\ -2 & -7 & -9 & 0 \end{array}\right]$$

$$\xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & -1 & -2 & -2 \\ 0 & -1 & -1 & 2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 3 & 4 & 1 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

↳ In echelon form.

Now, by Applying back substitution $x_3 = 4, x_2 = -6, x_1 = 3$.

So x = third col of A^{-1} = $\begin{bmatrix} 3 \\ -6 \\ 4 \end{bmatrix}$

ANS

★ Crammer's Rule: Let A^{-1} exist, then (Unique Soln)

$Ax = b \Rightarrow x = A^{-1}B$

$$x = \frac{1}{|A|} \begin{bmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

then if,

$$A_1 = \begin{bmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{bmatrix}, \quad A_2 = \begin{bmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{bmatrix}, \quad A_3 = \begin{bmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{bmatrix}$$

Just multiply cofactor & b

then, $x = \frac{1}{|A|} \begin{bmatrix} |A_1| \\ |A_2| \\ |A_3| \end{bmatrix}$ ✓

So crammer's rule is used to solve SOLE if A^{-1} exist. that is SOLE have an unique solution.
(But it is expensive as too much operations are involved, we prefer Back substitution as find A^{-1} & multiply by b & As Crammer's rule is expensive, it is also called Mathematics of Millionaires)

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Wishing you success on your journey ahead!

- Karan Agrawal